

Find the Square

Limits

⌚ Time

2000 ms

💾 Memory

1024 MiB

<> Source size

64 KiB

☑ Points

100

Submit

Code

C++

▼

Submit

Verdict	Time	Memory	Language	Points	Samples
compilation_error	0 ms	0 MiB	cpp	0/100	×
compilation_error	0 ms	0 MiB	cpp	0/100	×
compilation_error	0 ms	0 MiB	cpp	0/100	×
compilation_error	0 ms	0 MiB	cpp	0/100	×

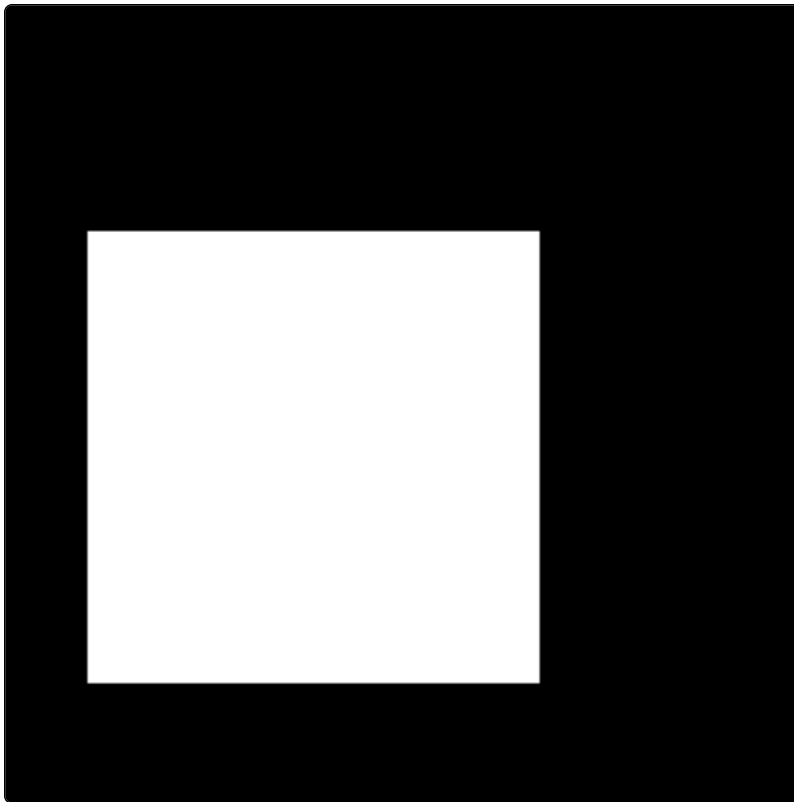
Description

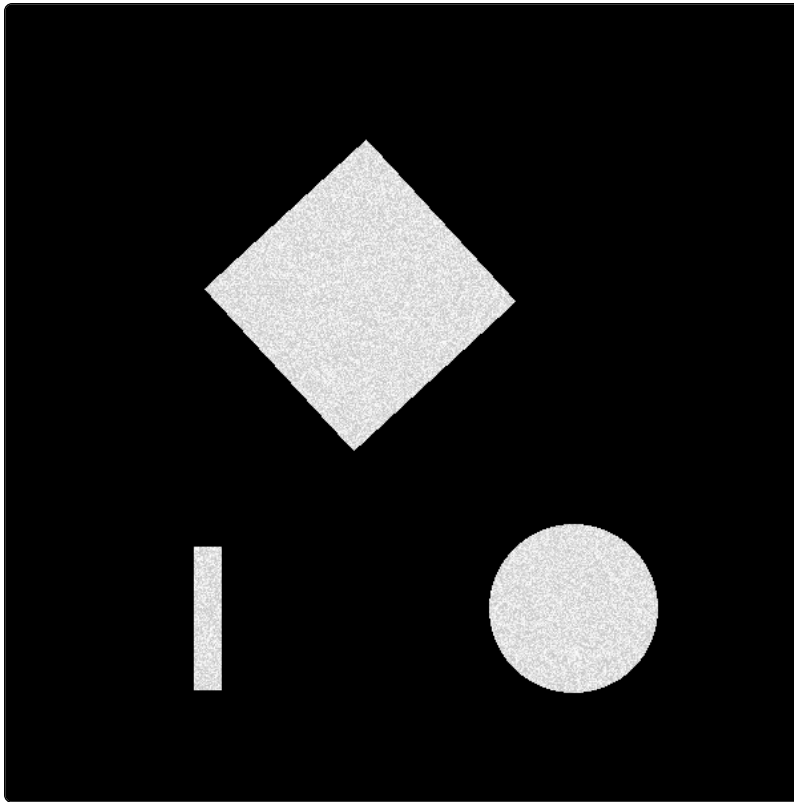
You are given a grayscale image of resolution $M \times N$. The rows and columns of the image are indexed from 0. That means the top row has index 0, the bottom row has index $M - 1$. The leftmost column has index 0 and the rightmost column has index $N - 1$. Every pixel is represented by an integer between 0 and 255. The value 0 indicates pure black and 255 indicates pure white and the values in between are various shades of gray.

It is known that somewhere on the image, there is a single white square which you need to find. Square is a rectangle with all sides of equal length. Its size and color can vary, it might be tilted, viewed from the side or the image might be noisy. It is guaranteed that the correct answer is undoubtedly clear when the image is viewed by a human.

To find the square, print the pixel coordinates (row first, then column) of each of its four corner pixels in the image. You can print them in any order and an error of ± 1 pixel in row and column will be tolerated.

Below you can see a few images, the likes of which will appear in the test set. You can see that some cases look very easy, others much more tricky.





There will be 50 test cases in total, each worth 2 points. The goal is not to score 100 points, as there is no single correct algorithm for this type of problem. Doing

simple things correctly is better than doing complicated things poorly. Given the time budget you have, try to cover as much cases as possible and score as high as possible.

Input Data

In the first line, there are two integers M, N in that order. Numbers M and N represent the number of rows and columns of our image and $50 \leq M, N \leq 1500$.

In each of the next M lines there are N integers $0 \leq x \leq 255$, the values of each pixel of the input image.

Output Data

Print 4 lines, one for each corner of the white square. In each line, print two integers, the coordinates of one of the corners of the targeted square. Print row coordinate first, then column coordinate. Remember that rows and columns are indexed from 0. Your output will be considered correct if all coordinates differ from the official (human) output by at most 1 pixel on each axis.

Scoring

There are 50 test cases, each worth 2 points. The cases will be of various sizes and difficulty.

Example

Input:

```
8 7
100 100 100 100 100 100 100
100 100 100 100 100 100 100
100 100 247 100 100 100 100
100 255 255 255 100 100 100
255 252 250 237 255 100 100
100 255 255 255 100 100 100
100 100 253 100 100 100 100
100 100 100 100 100 100 100
```

Output:

```
2 2
4 0
4 4
6 2
```

Explanation:

This is how approximately the zoomed-in image looks like. It is undoubtedly clear where the corners of the white square are located.

